

COURSE OUTLINE DETAILS

1. Course: Project - Basic Topics (Niên luận cơ sở ngành)

- **Code number:** CT239H
- **Credits:** 03
- **Hours:** 45 hours project.

2. Management Unit:

- **Department:** Software Engineering Department
- **Faculty/School/Institute/Center/Department:** College of Information and Communication Technology

3. Requisites:

- **Prerequisites:** **≥ 90 credits, CT174H**
- **Corequisites:** None

4. Course objectives:

Objective s	Descriptions	Program Outcomes
4.1	Show mastery of the basic theories of mathematics, sciences and technologies which are suitable for Software Engineering.	2.1.1b, 2.1.2a, 2.1.2b, 2.1.2c,
4.2	Proficiently use tools to design software solutions.	2.2.1c
4.3	Write documents properly and present results clearly and understandably.	2.2.2a
4.4	Demonstrate professionalism in software development activities	2.3a

5. Course learning outcomes:

CLOs	Descriptions	Objective s	POs
	Knowledge		
CLO1	Design algorithms and implement them in a specific programming language.	4.2	2.1.3a, 2.1.3b, 2.2.1a
CLO2	Write a scientific report clearly and accurately.	4.4	2.2.1b, 2.2.2a

CLOs	Descriptions	Objective s	POs
	Skills		
CLO3	Use proficiently tools to analyze real-life problems, thereby modeling problems, designing algorithms and programming solutions.	4.2	2.1.3a, 2.1.3b, 2.2.1a
CLO4	Plan to complete the project within time and resource constraints.	4.3	2.1.3c, 2.2.2b
CLO5	Writing a scientific report and presenting the project result.	4.4	2.2.2c
	Attitudes/Autonomy/Responsibilities		
CLO6	Have the spirit of learning to improve professional ability, proactively propose, be creative and dynamic at work.	4.5	2.3a

Note: CLO stands for Course Learning Outcome; PO stands for Program Outcome.

6. Brief description of the course:

This course helps students master the basic knowledge and skills for the Bachelor programme of engineering in software engineering. It helps the students to approach to small-scale problem solving completely by computers. The students will be assigned to develop an application project related to the knowledge of the subjects they have learned. They need to write a report and give a presentation of results.

During the development time, the students must regularly communicate with their instructor (via email or face-to-face) so that their instructor can monitor the project progress, solve problems (if any) and evaluate the results.

7. Course structure:

7.1. Theory

	Content	Hours	CLOs
1	Students self-review the knowledge they have learned related to their project		CLO3

7.2. Computer lab

	Content	Hours	CLOs
1	Planning, requirements analysis, design		CLO1
2	Implementation and validation		CLO1-CLO6
3	Write a scientific report		CLO2-CLO6
4	Product demo presentation		CLO2, CLO5

8. Teaching methods:

- Theory: The supervisor will provide or introduce necessary materials for students to read on their own.
- Practice: Students will perform the necessary work to achieve the project objectives.

9. Duties of student:

Students have to do the following duties:

- Meet the supervisor to determine the initial requirements at the beginning and during the project process.
- Perform work actively to accomplish the project objectives.
- Submit the report to the supervisor.
- Presenting results to the supervisor.

10. Assessment of course learning outcomes:

10.1. Assessment

No .	Point components	Rules and Requirements	Weights	CLOs
1	Ability of modeling, analyzing, and problem-solving	- Methods of problem-solving, suitable to the project topic. - Ability to apply the relevant knowledge learned.	20%	CLO1, CLO2, CLO3, CLO4, CLO6
2	Product demo program	The program needs to meet the requirements of the project topic	30%	All CLOs
3	The content of the report	The report is presented clearly and concisely	40%	All CLOs
4	Ability of project planning and ensuring project progress	Planning for the project and ensuring the project progress	10%	CLO3, CLO6

10.2. Grading

- Grading components and final test scores will be marked on a scale of 10 (0 to 10), rounded to one decimal place.
- Course score is the sum of all the components of the evaluation multiplied by the corresponding weight. The course score is marked on a scale of 10 and rounded to one decimal place, then it is converted to a letter score and a score on a scale of 4 under the academic regulations of the University.

11. Learning materials:

Learning materials information	Barcode number
Learning materials: books, textbooks, websites needed for student work will be introduced by instructors.	

12. Self-study Guide:

Week	Content	Theory (hours)	Practice (hours)	Student's Tasks
1	Identify the problem			Explore the root cause of the problem. Meet the supervisor to discuss about project requirements and get references.
2	Preliminary analysis			Explore all related solutions. Choose a possible solution to solve the problem. Discuss with the supervisor.
3	Planning and software requirements specification			Project planning Write software requirements specification
4-8	Development			Design, implementation, quality assurance and testing.
9	Write documentation			Write report.
10	Presentation			Submit the report, present the result, and run the program DEMO for supervisor's assessment.

**TL. HIỆU TRƯỞNG
TRƯỞNG KHOA**



Nguyễn Hữu Hòa

Cần Thơ, ngày 27 tháng 09 năm 2022

TRƯỞNG BỘ MÔN

Trương Minh Thái